

The allegory axioms, and what defines them in the Frobenius calculus

by

§1. Cartesian bicategory of relations

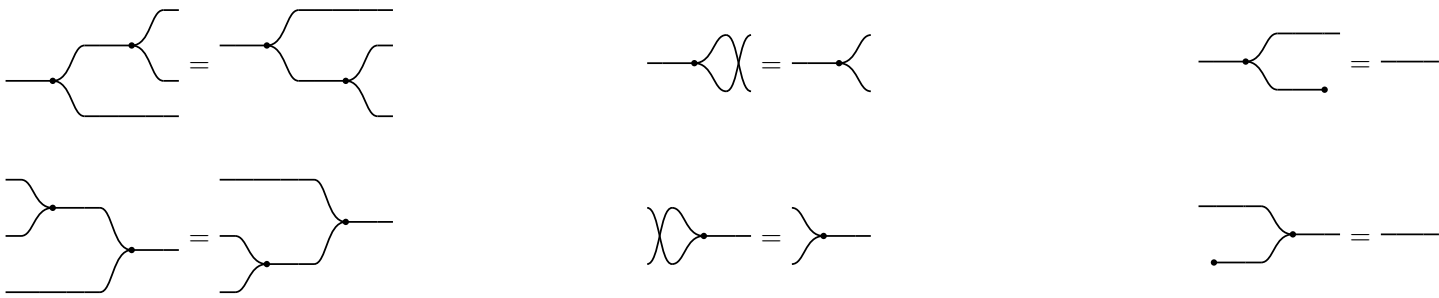
Definition 1.1.

A **cartesian bicategory of relations** is a poset-enriched category(bicategory) that is has a symmetric monoidal product $\otimes$ (cartisn product of set in Rel, vertical junctposition in the diagram) and

$\forall$ object A , $(A, \triangleleft, \multimap) \vdash (A, \triangleright, \multimap)$, **Frobenius**, arrows **lax**.

$(\triangleleft : A \rightarrow A \otimes A, \multimap : A \rightarrow \mathbb{I})$ formed a cocommutative comonoid

$(\triangleright : A \otimes A \rightarrow A, \multimap : \mathbb{I} \rightarrow A)$ formed a commutative monoid



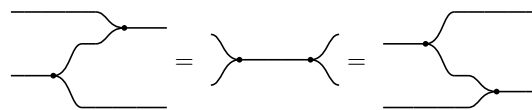
In $\text{Rel}(\text{Set})$, the four generators are these relations on a set A :

- $\triangleleft = \{(a, (a, a))\}$ — fork: one in, two copies out
- $\triangleright = \{((a, a), a)\}$ — merge: a pair passes only when its two components agree, and the common value comes out
- $\multimap = A \times \{*\}$ — discard: forget the value, keep only that there was one
- $\multimap = \{*\} \times A$ — create: any element at all

§1.1. $\forall$ object A , $(A, \triangleleft, \multimap) \vdash (A, \triangleright, \multimap)$



§1.2. The Frobenius law



The only clause coupling the comonoid to the monoid beyond adjointness. Both sides reduce to the same picture, $\triangleright \triangleleft$ — merge, then copy — which is the two-in two-out **spider**. Every connected diagram built from these four generators collapses to the spider on its own inputs and outputs, and this equation is what starts that collapse. A **bubble** is the other order, $\triangleleft \triangleright$: copy then merge, a closed loop, which the idempotency of n uses below.

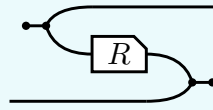
§1.3. Every arrow is a lax comonoid homomorphism



§2. $^\circ : \mathcal{C}^{\text{op}} \rightarrow \mathcal{C}$ is a 2 functor

Definition 2.1.

The **converse** R° of $R : a \rightarrow b$ is R with both of its wires turned round,

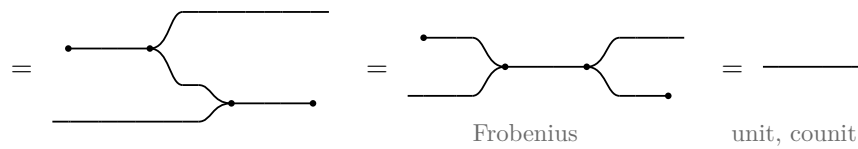


$$R^\circ = (\circlearrowleft \triangleleft \otimes 1) (1 \otimes R \otimes 1) (1 \otimes \triangleright \circlearrowright)$$

where $\circlearrowleft \triangleleft : \mathbb{I} \rightarrow a \otimes a$ opens a pair of wires out of nothing and $\triangleright \circlearrowright : b \otimes b \rightarrow \mathbb{I}$ closes one, so the input of R° is where the output of R was. And it is a contravariant 2-functor $^\circ : \mathcal{C}^{\text{op}} \rightarrow \mathcal{C}$:

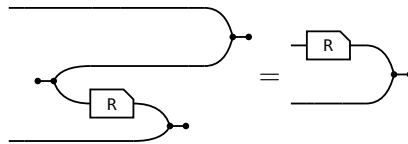
$$(i) \ 1^\circ = 1 \quad (ii) \ (R \ S)^\circ = S^\circ \ R^\circ \quad (iii) \ (R \otimes S)^\circ = R^\circ \otimes S^\circ \quad (iv) \ R \leq S \text{ implies } R^\circ \leq S^\circ$$

§2.1. $1^\circ = 1$ (snake)



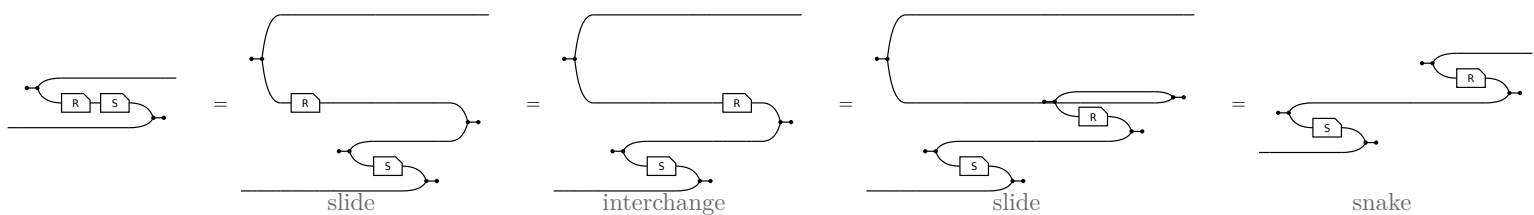
§2.2. The slide

The one rule (ii) and (iii) use, and each of them uses it twice. A converse facing a cap on the lower strand is the box itself, upright, on the upper one:



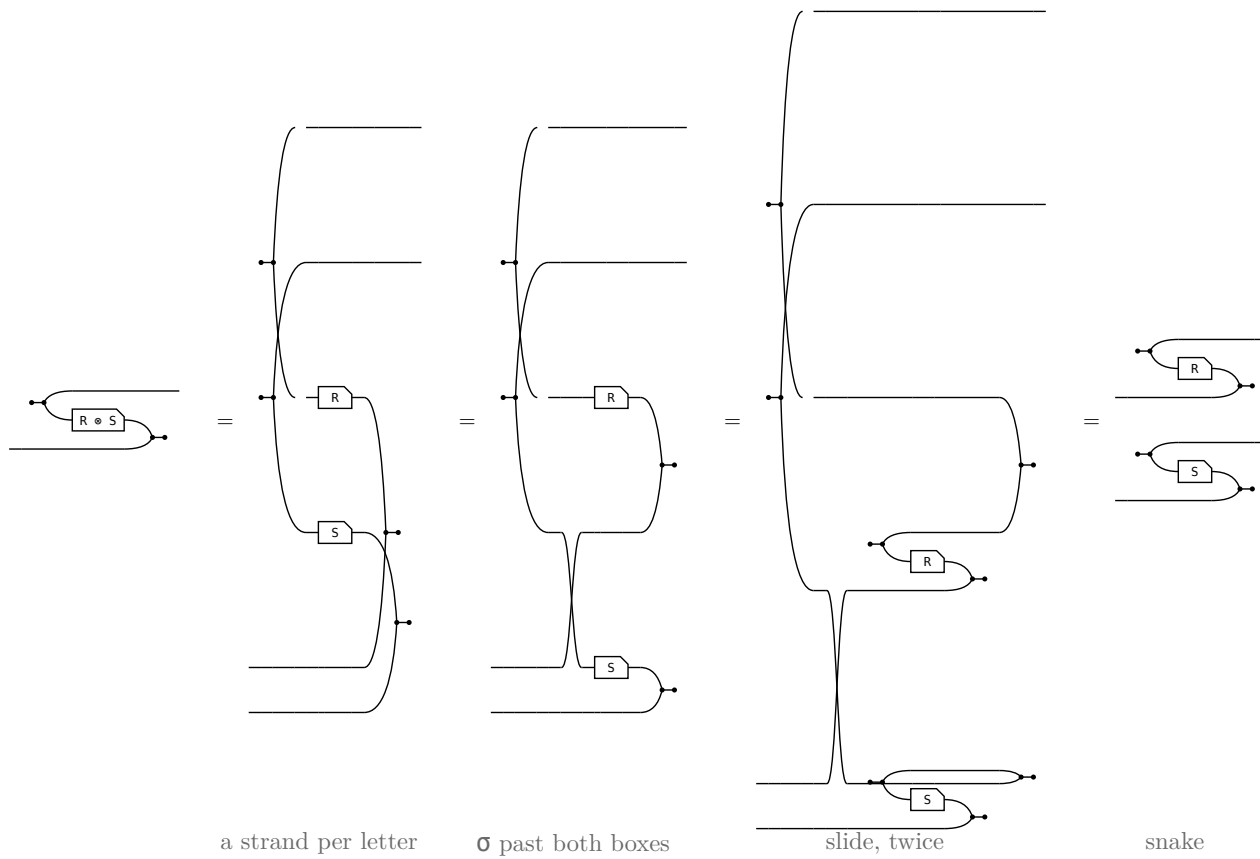
It is the snake again, read at R 's own bending: R° IS the bending of $(R \otimes 1)$ cap, so straightening it gives back what it was bent from. Nothing else is spent below.

§2.3. $(R \ S)^\circ = S^\circ \ R^\circ$



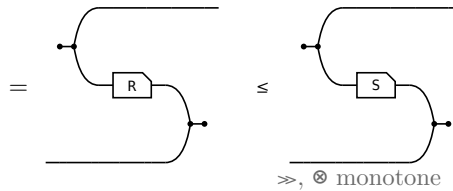
§2.4. $(R \otimes S)^\circ = R^\circ \otimes S^\circ$

A cap at a product is two caps behind a crossing, so the pair unbends one strand at a time and the crossing is all that is left to move.



§2.5. $R \leq S$ implies $R^\circ \leq S^\circ$

Here the cup IS drawn: R sits in a frame of wires built from $\gg$ and $\otimes$, and both of those are monotone, so the box may be replaced where it stands.



§3. Every arrow is a lax monoid homomorphism

The merge/create mirror of the two comonoid laws opening this note — the paper's (41) and (42), p. 22, which it gets by bending them with (ii) and (iv) above. No bending is needed: expand by the unit $1 \leq \triangleleft \triangleright$, duplicate the box with the comonoid law, contract by the counit $\triangleright \triangleleft \leq 1$. Both halves of the adjunction, and nothing else.

In $\mathbf{Rel}(\mathbf{Set})$ with $R \subseteq A \times B$, where $\triangleright$ is the equality test on a pair and $\bullet$ produces any element at all:

$$\begin{aligned}
 \triangleright R &= \{((a, a), b) : a R b\} \text{ — two equal inputs, then one run of } R \\
 (R \otimes R) \triangleright &= \{((a, a'), b) : a R b \text{ and } a' R b\} \text{ — a run on each input, then the results merged} \\
 \bullet R &= \{(*, b) : \exists a. a R b\} \text{ — the image of } R \\
 \bullet &= \{*\} \times B \text{ — all of } B
 \end{aligned}$$

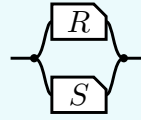
Take $a' = a$ for the first containment; every b lies in B for the second. Each is strict exactly when its reverse fails — R not injective, R not surjective — which is what the later table on maps reads off them.

inequation, and what proves it	picture
(41) $\triangleright R \leq (R \otimes R) \triangleright$ $\triangleright R = \triangleright R 1 \leq \triangleright R \triangleleft \triangleright \leq \triangleright \triangleleft (R \otimes R) \triangleright \leq (R \otimes R) \triangleright$ — unit, lax $\triangleleft$, counit.	
(42) $\bullet R \leq \bullet$ $\bullet R = \bullet R 1 \leq \bullet R \bullet \bullet \leq \bullet \bullet \bullet \leq \bullet$ — the same three steps at $\mathbb{I}$.	

§4. $\cap$ is a commutative idempotent monoid on every hom-set

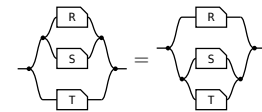
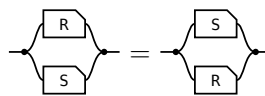
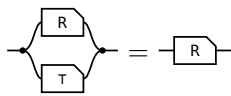
Definition 4.1.

The **meet** of $R, S : a \rightarrow b$, the paper's **convolution**, is $R \cap S := \triangleleft (R \otimes S) \triangleright$ — copy the input, run R and S on the two copies, merge the results — so what comes out is what both of them do.



transcribed: a definition has no statement to export

On every hom-set it is associative, commutative and idempotent, with unit the maximal arrow $\tau = \multimap \multimap$ (the paper's Lemma 4.11).

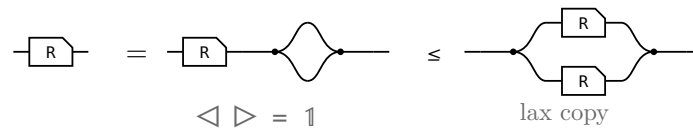


unit: one half of τ per end — the merge's unit law absorbs the $\multimap$, the copy's counit law the $\multimap$

commutative: σ crosses $R \otimes S$ by naturality and is absorbed by cocommutativity and commutativity

associative: coassociativity and associativity; $\otimes$ re-brackets for nothing, being strict here

§4.1. $R \cap R = R$, the one that is not bookkeeping



The last picture is $R \cap R$ itself, so this is $R \leq R \cap R$ and the lax copy law is the whole of it. The other direction is not proved here: $R \cap S \leq R$ holds for every S , by weakening S to τ and then collapsing $R \cap \tau$ — which is the paper's own remaining three steps, packaged once.

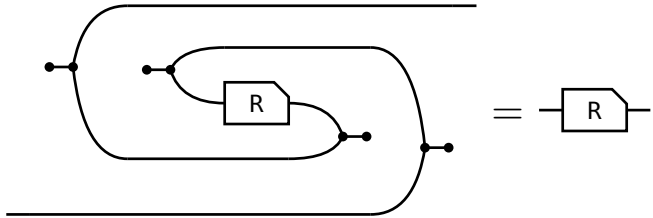
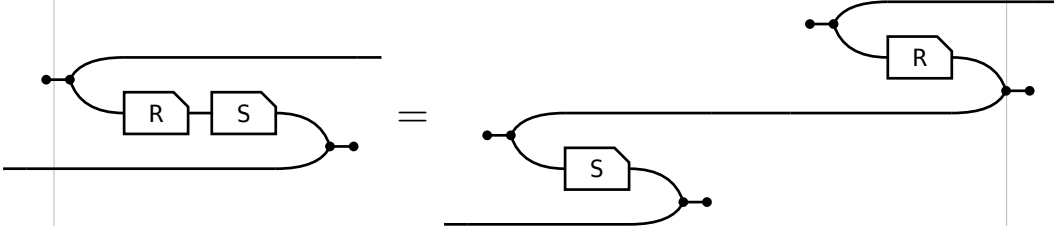
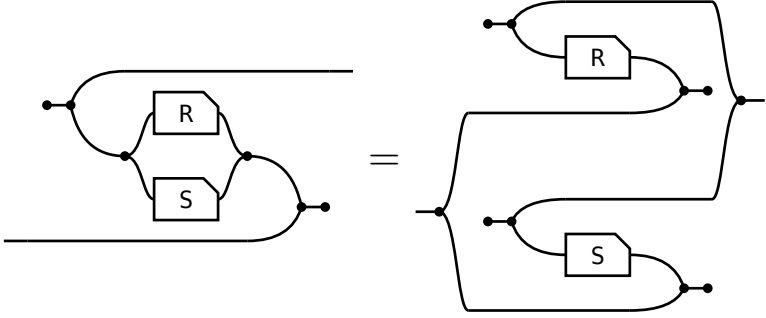
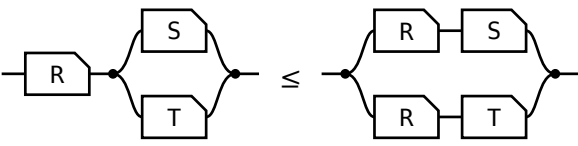
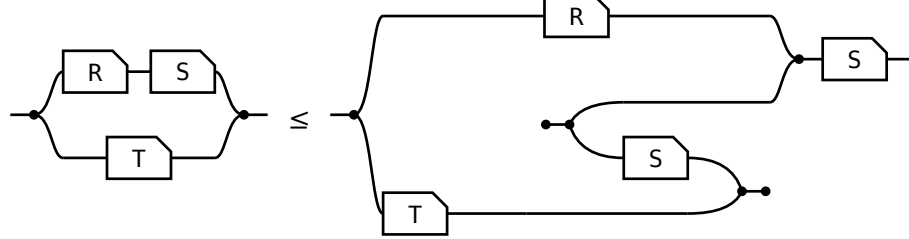
So $\leq$ is the order this monoid induces. $R \cap S \leq R$ comes from the unit, and idempotency turns anything under both S and T into something under $S \cap T$, since $R = R \cap R \leq S \cap T$.

§5. The allegory primitives are definitions here

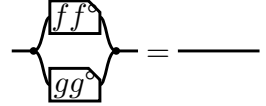
primitive, and what defines it here	picture
reciprocation $R^\circ := \text{bend } ((R \otimes 1) \text{ cap})$, not a generator.	
containment $R \sqsubseteq S$, which an allegory defines as $R \cap S = R$. Here $\leq$ is primitive and $\cap$ is derived, so the two directions are swapped.	the 2-cell itself
maximal morphism $\tau := \multimap \multimap$, free here; an allegory needs a unit object first.	

§6. The eight axioms, and what proves each

Three of the eight — $\cap$ idempotent, commutative, associative — are the monoid laws above. The other five:

axiom, and what supplies it	picture
$(R^\circ)^\circ = R$ — the snake .	
$(R \circ S)^\circ = S^\circ \circ R^\circ$ — mirroring the picture.	
$(R \sqcap S)^\circ = R^\circ \sqcap S^\circ$ — the same mirroring; $\triangleleft$ and $\triangleright$ are each other's mirror image.	
$R \circ (S \sqcap T) \sqsubseteq R \circ S \sqcap R \circ T$ — the lax copy law.	
$R \circ S \sqcap T \sqsubseteq (R \sqcap T \circ S^\circ) \circ S$, the modular law — adjoined as an axiom by an allegory, which has no structure to derive it from. Here Frobenius and the lax copy law give it.	

§7. Unitary and pre-tabular — what the other direction needs

condition	allegory	Frobenius side
unitary	$\mathbb{1}$ is a unit when $\mathbb{1} \circ T$ is the largest endomorphism on it and every object carries an entire arrow to it. A unit is what makes τ exist.	free. $\mathbb{1}$ is the unit object, $\mathbb{1} \leq \multimap \circ$ says $\multimap$ is entire, and $\tau = \multimap \circ$ is maximal.
pre-tabular	f, g tabulate R when both are maps, $R = f \circ g$, and  R is tabular when some pair tabulates it, the allegory pre-tabular when every arrow lies under a tabular one.	this is what $\otimes$ is. Given unitarity it reduces to tabulating each τ , and a tabulation of τ is an object $n \otimes m$ with its two projections. One side derives the product from tabulations, the other posits it.

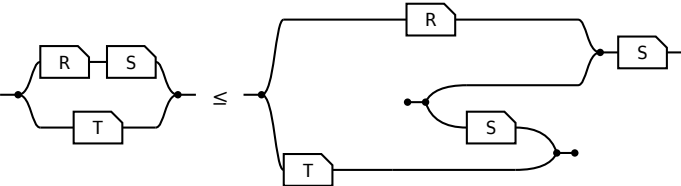
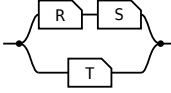
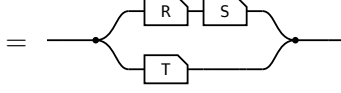
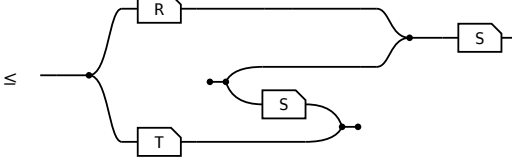
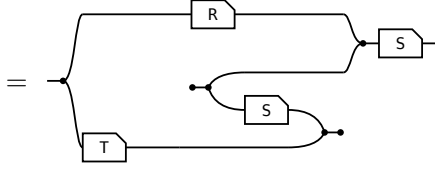
Together: cartesian bicategory of relations $\simeq$ unitary pre-tabular allegory.

§8. Union and residual

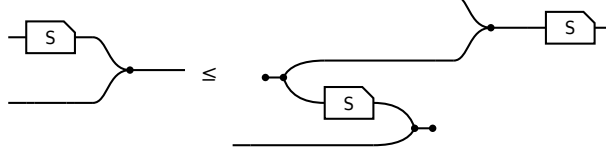
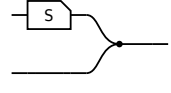
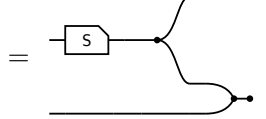
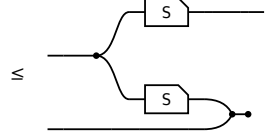
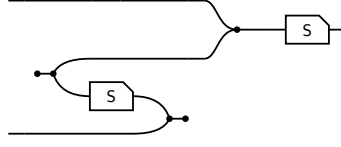
Both are past what the definition can build: a union needs a **biproduct** on top of the Frobenius structure, a residual a **second composition** with linear adjoints. A union draws as a **tape** — the rounded wrapper is the second product, and its fork and join open and close a branch a particle takes exactly one of. A residual draws as **long division**, not as its own term, which is a composition against a complement.

statement	picture
$R \sqsubseteq R \cup S$ union $R S := \langle R, S \rangle [1, 1]$: offer both branches, then forget which was taken.	
$R \cup S = S \cup R$	
$\perp \cup R = R$ $\perp$ routes through the zero object; there is no shape for it.	
$R (S \cup T) = R S \cup R T$ — composition distributes over union, which n does not.	
$(R \cup S)^\circ = R^\circ \cup S^\circ$	
$R \cap (S \cup T) = (R \cap S) \cup (R \cap T)$ — what makes the whole distributive , and the one picture with both layers in it.	
$(R / S) S \sqsubseteq R$ — the residual is a lower bound of the arrows it is the greatest of.	

§9. The modular law, from Frobenius

the modular law, $R \ S \cap T \leq (R \cap T \ S^\circ) \ S$ 		
term	picture	the rule that reaches it
$(R \ S) \cap T$		the start: $R \ S \cap T$.
$= (\triangleleft (R \otimes T)) ((S \otimes 1) \triangleright)$		Factor out the shared prefix: $(R \ S) \otimes T$ is $(R \otimes T) (S \otimes 1)$.
$\leq (\triangleleft (R \otimes T)) ((1 \otimes S^\circ) (\triangleright S))$		The merge slide. S slides through the merge and reappears as S° on the other strand. The only inequality in the proof.
$= (R \cap (T \ S^\circ)) \ S$		Fold the prefix back in; the copy-merge pair reads as a meet again.

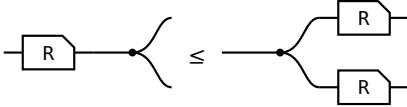
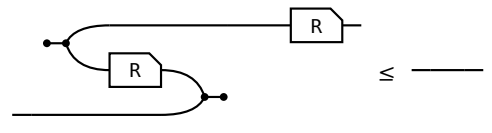
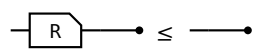
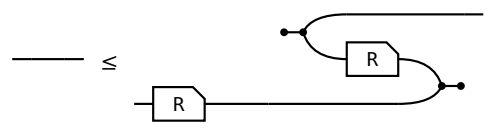
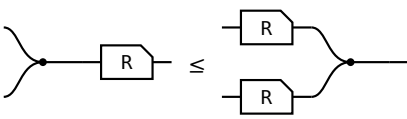
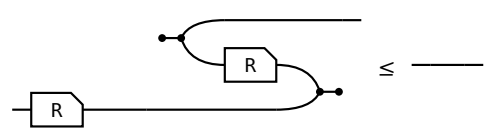
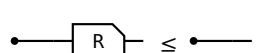
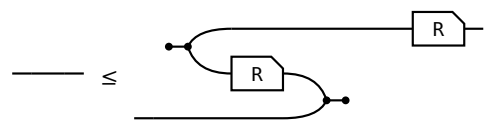
S occurs once on the left of the slide and twice on the right, and the lax copy law is the only law in the definition that may duplicate a box — so that is where the inequality has to be.

the merge slide, $(S \otimes 1) \triangleright \leq (1 \otimes S^\circ) \triangleright S$ 		
term	picture	the rule that reaches it
$(S \otimes 1) \triangleright$		the start: $(S \otimes 1) \triangleright$.
$= ((S \triangleleft) \otimes 1) (1 \otimes \text{cap})$		Rebuild the merge from a copy and a right bracket, the shape the lax copy law applies to.
$\leq ((\triangleleft (S \otimes S)) \otimes 1) (1 \otimes \text{cap})$		The lax copy law: $S \triangleleft$ becomes $\triangleleft (S \otimes S)$, the box copied. The whole of the modular law, in one step.
$= (1 \otimes S^\circ) (\triangleright S)$		Reshape back, and the surviving duplicate slides round the right bracket to become S° .

§10. Maps

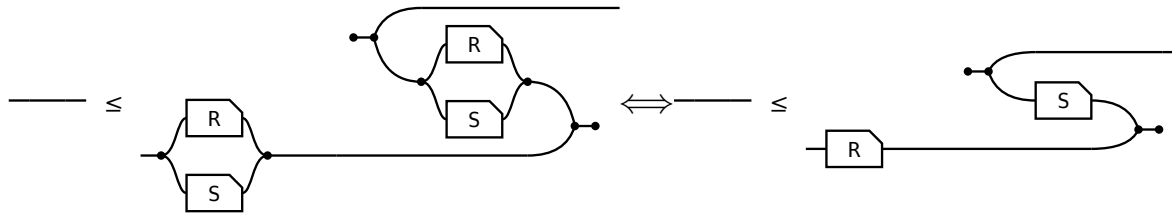
Every arrow is lax for $\triangleleft$ and for $\multimap$ by axiom, and for $\triangleright$ and $\multimap$ by the section above. All four hold for **every** arrow, their REVERSES do not, and each reverse holding is a property of the arrow. The right-hand column is the **adjoint** form, which

is the definition used here because it is literally the allegory's **Simple** and **Entire**; that the two forms agree is a separate theorem, not proved here.

holds for every arrow	property	holds iff
$R \triangleleft \leq \triangleleft (R \otimes R)$ 	(SV) single valued $R^\circ R \sqsubseteq 1$	
$R \multimap \leq \multimap$ 	(TOT) total $1 \sqsubseteq R R^\circ$. With single valued, a map.	
$\triangleright R \leq (R \otimes R) \triangleright$ 	(INJ) injective $R R^\circ \sqsubseteq 1$	
$\multimap R \leq \multimap$ 	(SUR) surjective $1 \sqsubseteq R^\circ R$, that is, R° entire.	

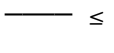
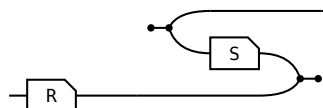
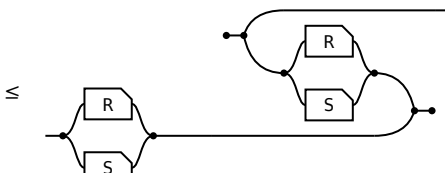
§11. Entireness of an intersection

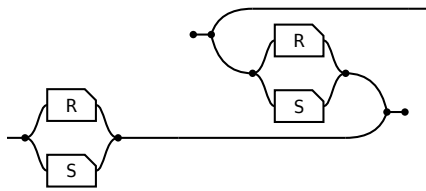
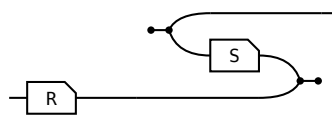
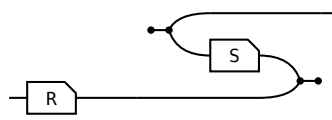
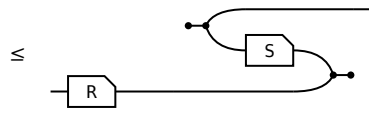
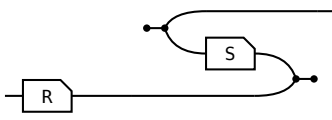
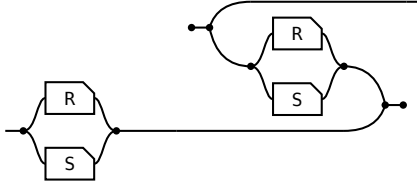
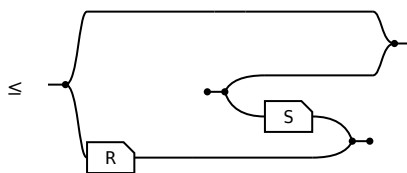
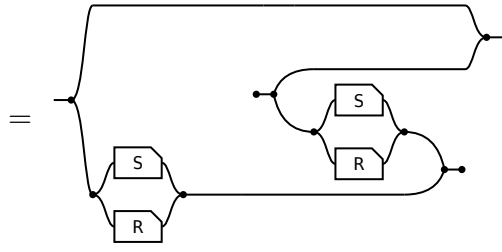
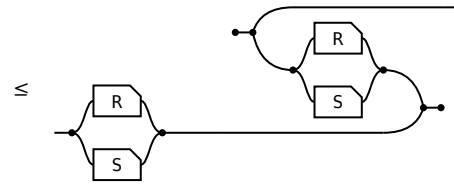
R is **entire** when $1 \sqsubseteq R R^\circ$, which is (TOT) above. When is an **intersection** entire?



$$\text{Total } (R \cap S) \leftrightarrow 1 \leq R S^\circ$$

On the left $R \cap S$ is named **twice**; on the right R and S once each and the meet is gone. The meet is needed to **ask** whether something is entire, not to **answer** it. Where $\sqsubseteq$ is **defined** as $X \cap Y = X$ the two sides are one proposition; here $\leq$ is primitive, so both directions are real steps — and they cost very different things.

$\Rightarrow$ given $R \cap S$ entire,  show 		
term	picture	the rule that reaches it
1		the start: 1.
$\leq (R \cap S) (R \cap S)^\circ$		the hypothesis: $1 \leq P P^\circ$ at $P = R \cap S$.

$\Rightarrow$ given $R \cap S$ entire,  $\leq$  show $\leq$ 		
term	picture	the rule that reaches it
$\leq R S^\circ$		monotonicity, twice. No modular law.
$\Leftarrow$ given  $\leq$ show $R \cap S$ entire, $\leq$ 		
term	picture	the rule that reaches it
1		the start: 1.
$\leq 1 \cap (R S^\circ)$		the hypothesis, met with 1.
$= 1 \cap ((S \cap R) (S \cap R)^\circ)$		$1 \cap R S^\circ = 1 \cap (S \cap R) (S \cap R)^\circ$. $R S^\circ$ relates a to a' when some b has $a R b$ and $a' S b$; the $1 \cap$ sets $a' = a$, and it reads “some b has $a R b$ and $a S b$ ” — one arrow of $R \cap S$. Two arrows become one, and that is what needs the modular law.
$\leq (R \cap S) (R \cap S)^\circ$		$X \cap Y \leq Y$, then $S \cap R = R \cap S$.

§12. The gaps

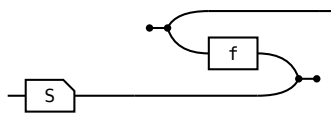
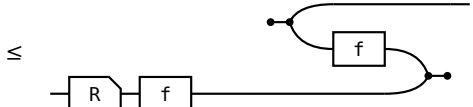
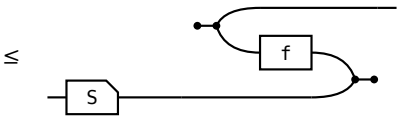
- All eight axioms are theorems of the definition, the modular identity included.
- $\triangleright \triangleleft \leq 1$ and $\circ \circ \leq 1$ are never used. They are there to make $\triangleleft \dashv \triangleright$ and $\circ \dashv \circ$ genuine adjunctions, which is what pins $\triangleright = \triangleleft^\circ$ and $\circ = \circ^\circ$.
- Unproved: the merge and unit forms of the lax laws, and the agreement of the comonoid and adjoint forms of the map conditions. Both trace to the one omission above.
- The bridge runs one way only: a cartesian bicategory carries $\otimes$, an allegory carries nothing of the kind, which is what the two conditions supply.
- Everything here is a theorem OF the axioms. That relations satisfy them is not checked here.

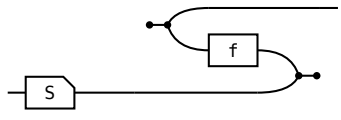
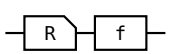
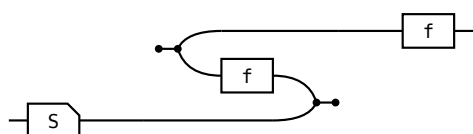
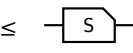
construct	counterpart	status
tabulation	none	$\tau = \circ \circ$ is all the tabular content there is.
$\cup, \perp$	a biproduct $\oplus$ on top of the Frobenius structure	built, and drawn above. The calculus itself has no union.

construct	counterpart	status
R / S , complement	a second composition with linear adjoints	built, and drawn above, but axiomatised only: nothing instantiates the complement.
Λ , power transpose	none	open. No power object anywhere in this tower.
allegory $\Rightarrow$ cartesian bicategory	rebuild $\otimes$ from tabulations	deliberately not built.

§13. The shunting rule

A map moves from one side of a containment to the other at the cost of its converse, $R \cdot f \leq S \iff R \leq S \cdot f^\circ$. It is how a function gets out of the way so the relation underneath can be reasoned about. Each direction spends exactly one half of **map** — never both.

shunting right, $\Rightarrow$ given  $\leq$  show  $\leq$ 		
term	picture	the rule that reaches it
R		the start: R .
$\leq R \cdot (f \cdot f^\circ)$		total: $1 \leq f \cdot f^\circ$, so $f \cdot f^\circ$ may be inserted after R .
$\leq S \cdot f^\circ$		Re-bracket to $(R \cdot f) \cdot f^\circ$ and apply the hypothesis.

shunting right, $\Leftarrow$ given  $\leq$  show  $\leq$ 		
term	picture	the rule that reaches it
$R \cdot f$		the start: $R \cdot f$.
$\leq S \cdot (f^\circ \cdot f)$		Apply the hypothesis on the left; the two f s are now adjacent.
$\leq S$		single valued: $f^\circ \cdot f \leq 1$, so the pair cancels.

Shunting left is the mirror, same shape, map on the other side throughout. What the rule is, is an adjunction: $f \dashv f^\circ$, with **total** the unit and **single valued** the counit, so “ f is a map” and “ f has a right adjoint, namely f° ” are one statement.

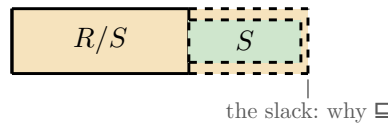
§14. Division

Division is drawn and negation is not, because negation is not how programs get specified: the chapters where they are — greedy, thinning, dynamic programming, Horner, knapsack, paragraph formatting, string edit, bracketing, compression — use local completeness, division and power objects, and no Boolean structure at all. That also settles which picture of $/$ is right. As a term a residual is a composition against a complement, which makes every homset Boolean; division asks for

nothing of the kind, only the Galois connection $T \sqsubseteq R/S \iff T S \sqsubseteq R$, which $\mathbf{Rel}(\mathbf{Set})$ satisfies with a bare $\forall$, $(R / S) \times y \triangleq \forall z. S y z \rightarrow R \times z$.

Long division. R / S is how much of R you can lay down before S still fits in what is left. Read the bar as a composite: R / S first, then S , the two together inside R . Solid is what the equation pins — R 's left edge, S 's left edge, and the stretch before S . Dashed is the slack: nothing to the right of S is determined by $T S \sqsubseteq R$, which is why the cancel law is $(R/S) S \sqsubseteq R$ and not an equality, and why the slack is a hairline — R / S is the **largest** such quotient.

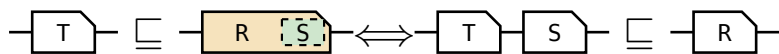
It is a **measurement**, not a containment: $R : a \rightarrow c$ and $S : b \rightarrow c$ are not in the same hom-set, so what sits inside R is the composite $(R/S) S$. Pushed too far the metaphor breaks in the usual place, $\perp / \perp = \top$.



transcribed — the universal property has a statement, this metaphor does not

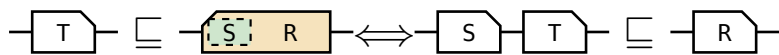
the ten laws, their pictures, and — where the metaphor reaches — the same law as long division

Division is the right adjoint of composition: $T \sqsubseteq R/S \iff T S \sqsubseteq R$



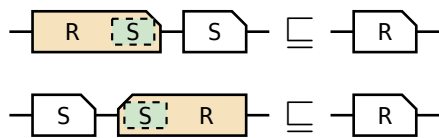
$T \sqsubseteq S \setminus R \iff S T \sqsubseteq R$

$S \setminus R \triangleq (R^\circ / S^\circ)^\circ$: left division is right division conjugated by the converse, not a second primitive.



cancel: $(R/S) S \sqsubseteq R$, and $S (S \setminus R) \sqsubseteq R$

The counit of each adjunction — the law the figure above draws.



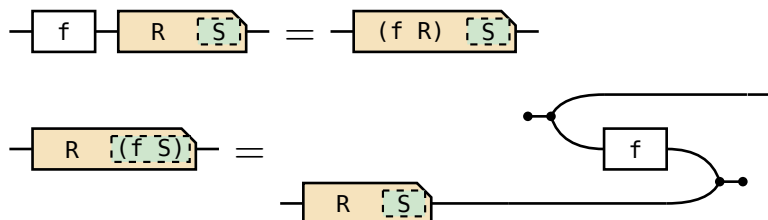
associate: $R / (S_1 S_2) = (R / S_2) / S_1$, and $(S T) \setminus R = T \setminus (S \setminus R)$

Dividing by a composite one factor at a time; the bar shows what the label on the right hides — two tiles laid where there was one.



maps: $f (R/S) = (f R) / S$, and $R / (f S) = (R/S) f^\circ$

The same three pieces bracketed two ways, which is the licence to write $f R / S$; then a map moves out of a denominator and reappears as f° outside the box. Both are the shunting rule, spent twice.



symmetric division: $(R /_s S)^\circ = S /_s R$, and $(R /_s S) (S /_s T) \sqsubseteq R /_s T$

$R /_s S \triangleq (R/S) \cap (S/R)^\circ$. The meet is inside the definition, not the statement, so it stays under the label: $/_s$ is converse-symmetric and transitive, like an equality of columns.

the ten laws, their pictures, and — where the metaphor reaches — the same law as long division

$$\begin{array}{c}
 \begin{array}{c} \bullet \\ \text{---} \end{array} \begin{array}{c} \text{---} \\ \text{---} \end{array} \begin{array}{c} \bullet \\ \text{---} \end{array} \\
 \text{---} \begin{array}{c} \text{---} \\ \text{---} \end{array} \bullet \\
 \end{array} \begin{array}{c} \boxed{R /_s S} \end{array} = \begin{array}{c} \boxed{S /_s R} \end{array}$$

$$\begin{array}{c} \text{---} \end{array} \begin{array}{c} \boxed{R /_s S} \end{array} \begin{array}{c} \boxed{S /_s T} \end{array} \begin{array}{c} \text{---} \end{array} \sqsubseteq \begin{array}{c} \boxed{R /_s T} \end{array}$$

Ten laws, ten pictures, and not one shows a generator: $\mathfrak{n}$, $\mathfrak{u}$, $^\circ$ and composition are what the Frobenius generators build, and $/$ is none of those — it is posited, with nothing to unfold.